

Problem 1

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1.a : 5 states of processes

- ▣ New
- ▣ Ready
- ▣ Running
- ▣ Waiting (Blocked)
- ▣ Terminated

1.b Running State

Problem 2

CPU is in sec not m/s

0.01 m/s

0.1 m/s

200 min

50 m/s

1 second = 1000 m/s

$$\frac{200}{60} = 3.333 \text{ sec}$$

2.a $3.333 \times 0.01 = 0.03333 \text{ m/s}$

$$0.0333 / 1000 = 0.0000333 \text{ sec}$$

$$\left(\frac{0.0000333}{1} \right) \times 100 = \boxed{0.00333\% \text{ CPU Time}}$$

$$\frac{1000}{50} = 20 \text{ interrupts per sec}$$

$$\rightarrow 20 \times 0.1 = 2 \text{ m/s}$$

$$\frac{2}{1000} = 0.002$$

$$\frac{0.002}{1} \times 100 = \boxed{0.2\% \text{ CPU Time}}$$

2.b 0.0000333% CPU

vs.

0.2% CPU

so scenario 1 use less CPU Time so
executing task is faster

2.c 6.2% CPU Time per interrupt

so Scenario 2 has less interrupts
per second which improve CPU utilization

Problem 3

10 Mbps

10 packets every 50 ms

↓

1000 bytes

$1000 \times 8 = 8000 \text{ bits}$

1 Mbps = 1,000,000 bits

3.a 1 byte = 8 bits

3.b $\frac{10}{50} = 0.2 \rightarrow 0.2 \times 1000 = \boxed{200 \text{ packages per sec}}$ ^{Network card}

pkg $\rightarrow$ bits

$200 \times 8000 =$
 $1,600,000$

$\frac{1600000}{1000000}$

$\boxed{1.6 \text{ Mbps}}$

3.c Utilization = $\frac{\text{CPU Time Used}}{\text{Total CPU Time}} \times 100$

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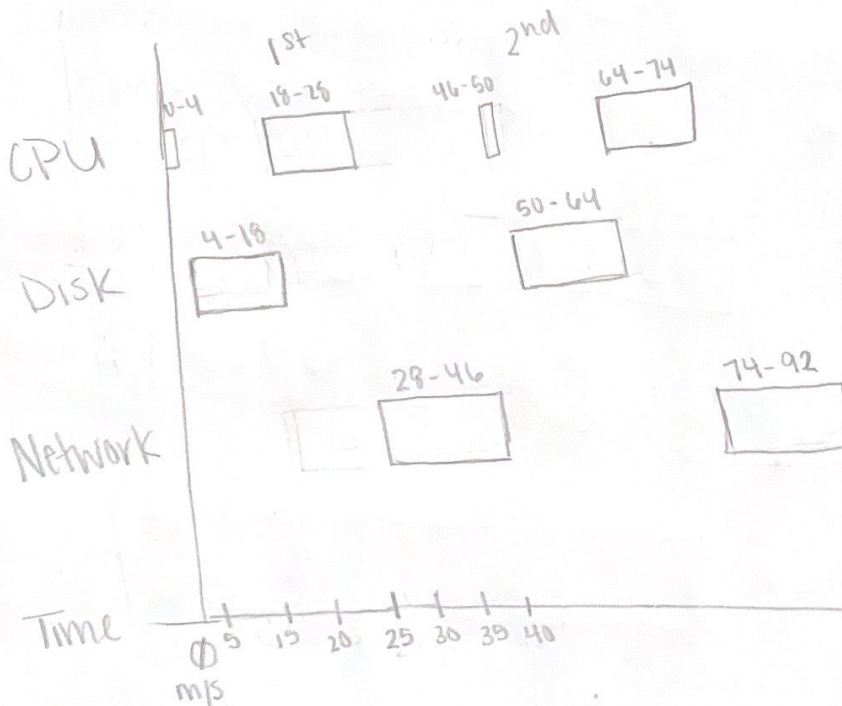
$$\frac{1.6 \text{ Mbps}}{10 \text{ Mbps}} \times 100 = \boxed{16 \% \text{ Utilization}}$$

Problem 4

- CPU = 4 mps $4 + 10 + 4 + 10 = 28$
- Disk = 14 mps $14 + 14 = 28$
- CPU = 10 mps
- Network = 18 mps $18 + 18 = 36$

92 = Total Time

4.a



$92/2 = 46$
each iteration

4.b $46 \text{ ms} \times 2 \text{ iterations} = 92 \text{ ms}$

CPU utilization $\frac{28}{92} \approx 0.3043 \times 100 = \boxed{30.43\%}$

Disk Utilization $\frac{28}{92} \approx 0.3043 \times 100 = \boxed{30.43\%}$

Network Utilization $\frac{36}{92} \approx 0.3913 = \boxed{39.13\%}$

4.c.d

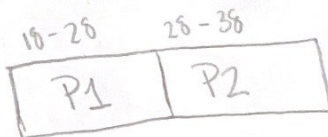
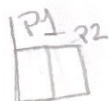
P1 starts @ 0 ms
then 0-4 ms

P2 starts @ 4 ms
4-8 ms

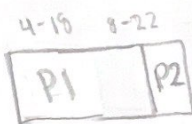
Alternates between the two independent processes until both processes complete their steps

No Overlap

CPU



$\frac{28}{92} = 0.5 \times 100 = \boxed{30.43\%}$

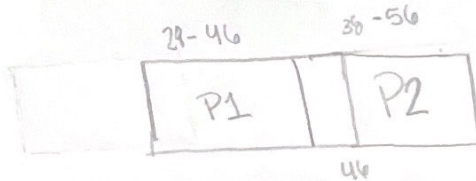


$\frac{28}{92} \times 100 = \boxed{30.43\% \text{ Disk}}$

Overlap

Disk
 $14+14=28$

Network
 $18+18=36$



$\frac{36}{92} \times 100 = \boxed{39.13\% \text{ Network}}$

Time

$46 \times 2 = 92$

Problem 5

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A $0+2=2$

B $2+3=5$ $3 \rightarrow 7$ gap

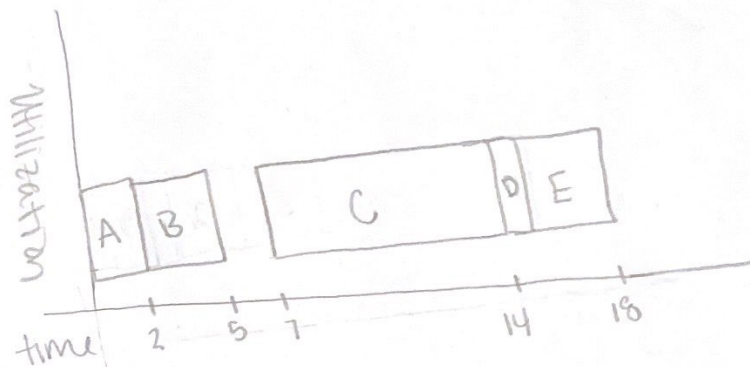
C $1+6=13$

D $13+1=14$

E $14+4=18$

5.a 18 seconds till all processes done

5.b



5.c $A(2) + B(3) + C(6) + D(1) + E(4) = 16$

$\frac{16}{18} \times 100 \approx \boxed{88.89\%}$ average utilization

5.d $\frac{5 \text{ processes}}{18 \text{ CPU}} \approx \boxed{0.278\% \text{ processors/s}}$

5.e turnaround = time b/t arrive + finish

Average Turnaround $\frac{26}{5} = \boxed{5.2 \text{ sec.}}$

$2-0=2$
 $5-1=4$
 $13-7=6$
 $14-8=6$
 $18-10=8$

 26

Problem 6

month = 30 days

$$30 \times 24 = 720 \text{ hrs}$$

A 2 hrs a mon 1 time

B 20 min 6 times a mon. $6 \times \frac{1}{3} = \underline{2 \text{ hrs}}$

G.a $\frac{720 - 2}{720} \times 100 = \boxed{99.72\%}$ for A availability

G.b $\frac{720 - 2}{720} \times 100 = \boxed{99.72\%}$ for B availability

G.c MTBF of system A

$$\frac{\text{Total Time}}{\# \text{ of Failures}} = \frac{720}{1} = \boxed{720 \text{ hr}}$$

longer downtimes

G.d MTBF of system B

$$\frac{720}{6} = \boxed{120 \text{ hrs}}$$

shorter downtimes but more frequent